Name: PM Meenakshi

SPATIAL MAPPING AND ANALYTICS OF WASTE GENERATION AND PARTICIPATION

Using Data-Driven Insights to Understand Urban Waste Patterns

Mail: meenakshipmurali@gmail.com

Contact :9489683941

TOOLS AND LIBRARIES USED

- Python – For all data processing and visualization tasks
- Pandas – To handle, clean, and aggregate data
- Matplotlib & Seaborn – For creating time series, bar charts, and scatter plots
- Folium – To develop interactive map visualizations
- GitHub & GitHack – For hosting and sharing HTML-based interactive maps
- Canva – To design and organize the final presentation
- Jupyter Notebook – As the working environment for code and output
- ChatGPT – Used for brainstorming ideas, validating approaches, and exploring data visualization options
- External Data Sources – Included an additional pincode dataset (for geolocation) and population data to support deeper analysis

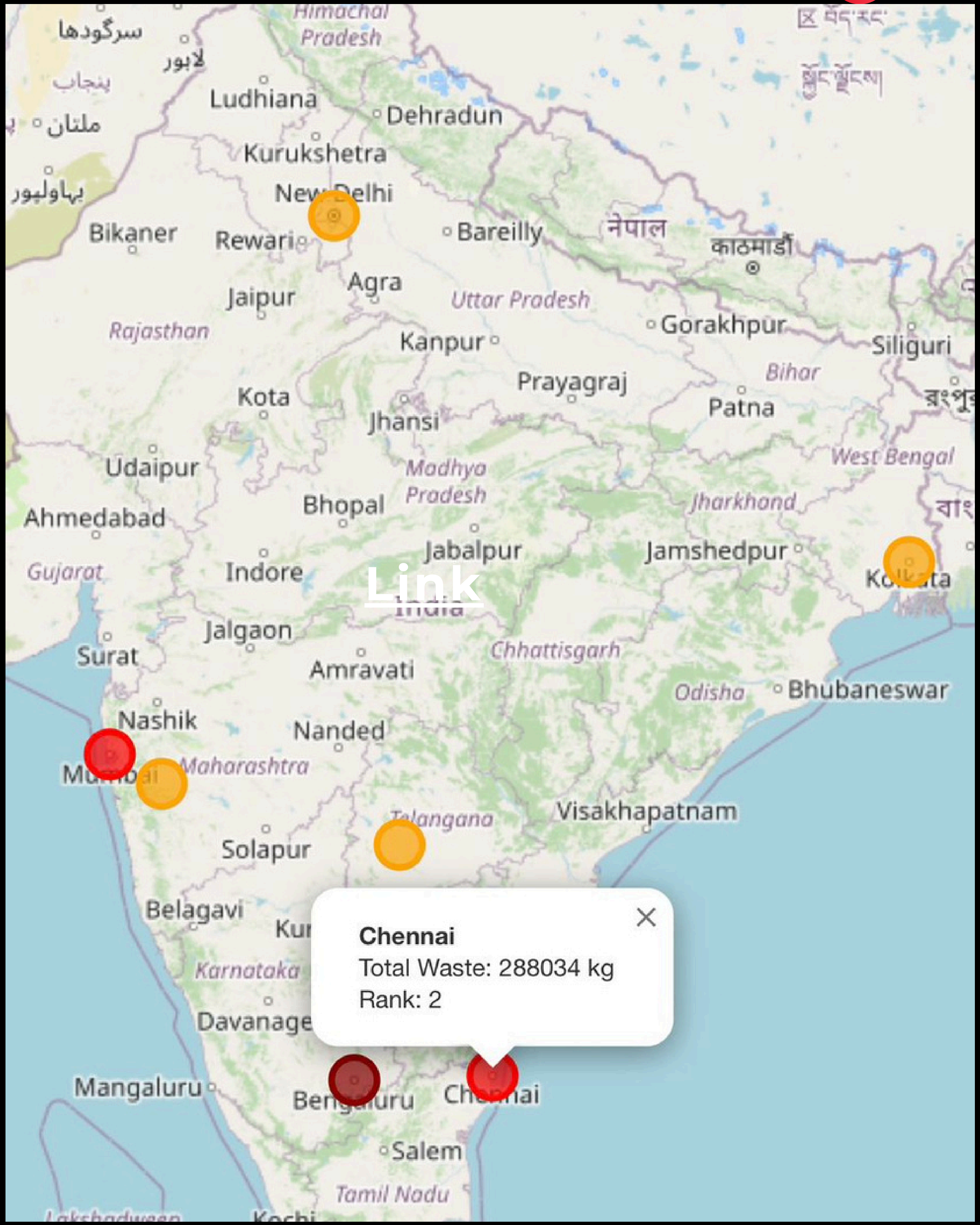
INTERACTIVE HEATMAP - WASTE GENERATION BY CITY

To view the interactive heatmap of waste generation by city, click on the link - ***Link***

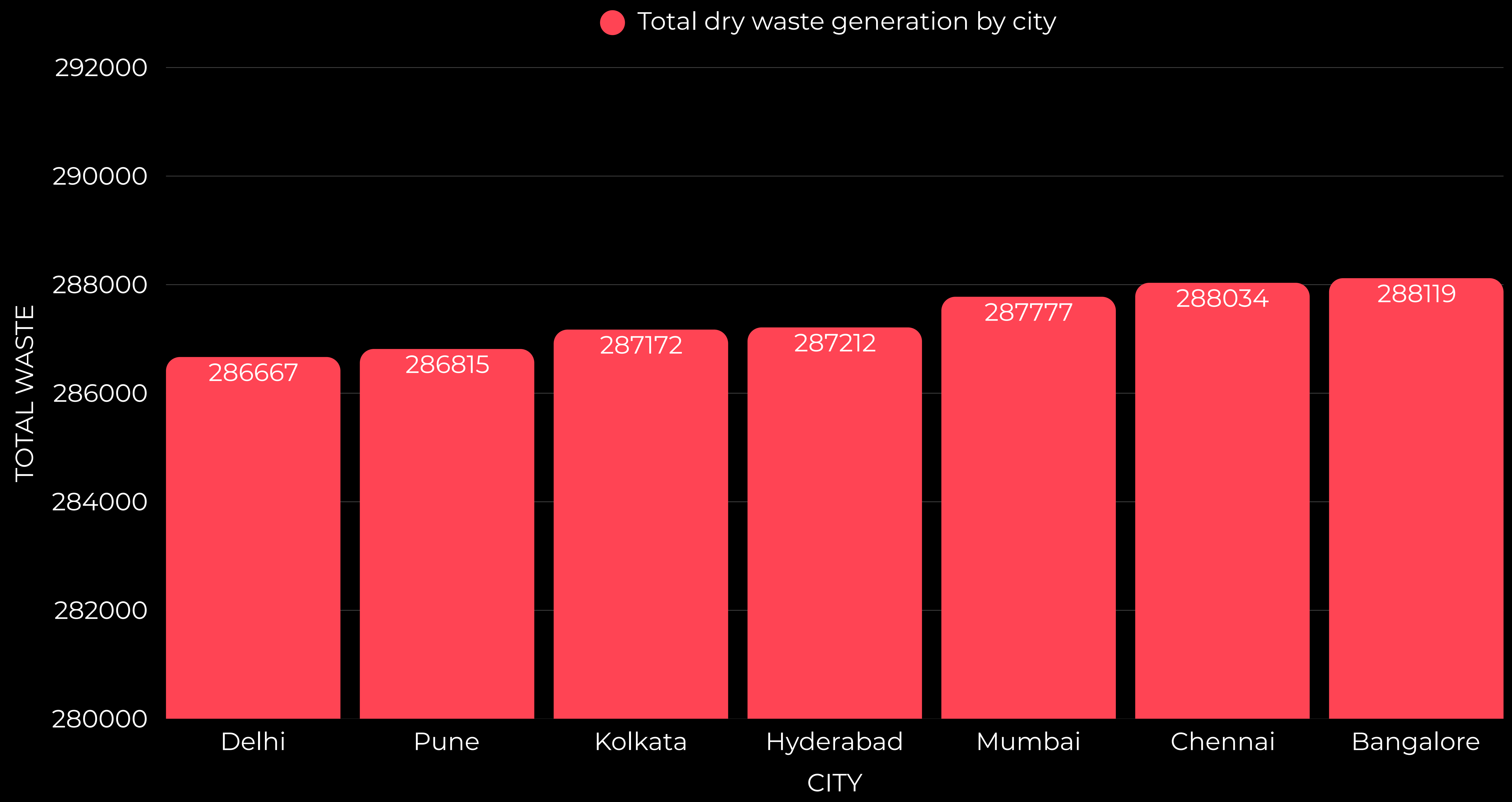
This heatmap visualizes the total dry waste generated by major cities across India using a spatial mapping tool.

What it shows:

- Color intensity represents the volume of dry waste generated.
- Cities like **Bangalore, Mumbai, and Chennai** emerge as **high-waste zones**, with dark red markers indicating the top ranks.
- Data is aggregated at the city level, using **geospatial coordinates.**



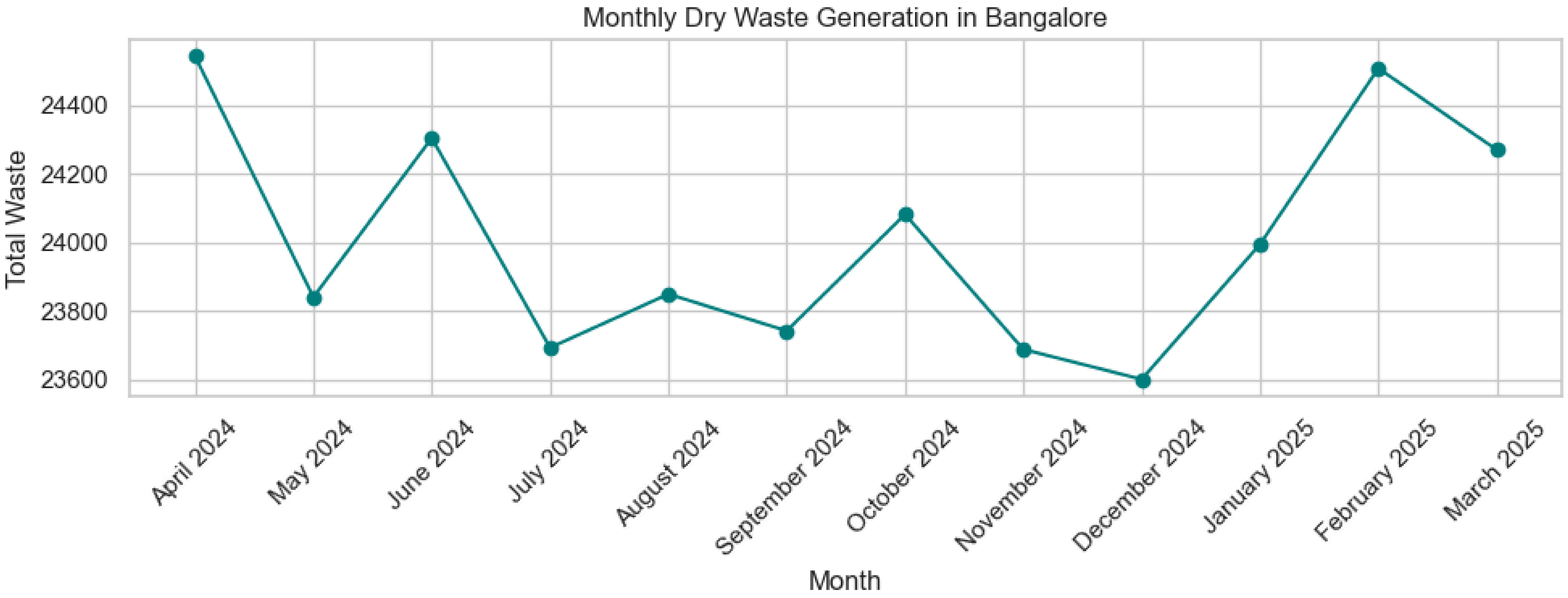
BAR PLOT OF TOTAL WASTE GENERATED FOR EACH CITY



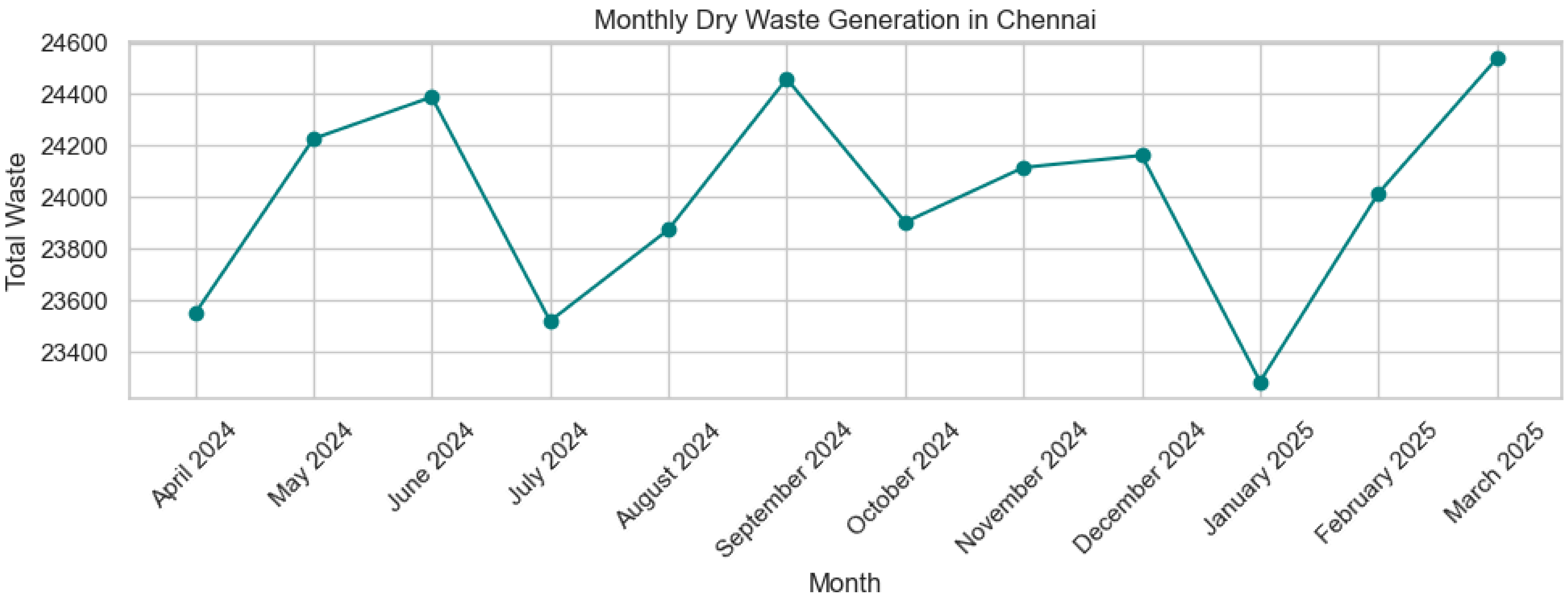
INFERENCE FROM THE PREVIOS GRAPH

- **Bangalore** records the highest dry waste generation at **288,119 kg**.
- **Chennai** and **Mumbai** follow closely, with only marginal differences.
- **Delhi** and **Pune** are on the **lower** end among the selected cities.
- The variation in waste generation across all seven cities is less than **1%**, showing consistent levels.
- This consistency likely stems from similar **population density**, **consumption habits**, and **waste collection systems.**
- Slight differences may reflect **segregation efficiency, reporting practices, or infrastructure disparities.**

MONTHLY TRENDS IN CITY-WISE WASTE GENERATION BANGALORE

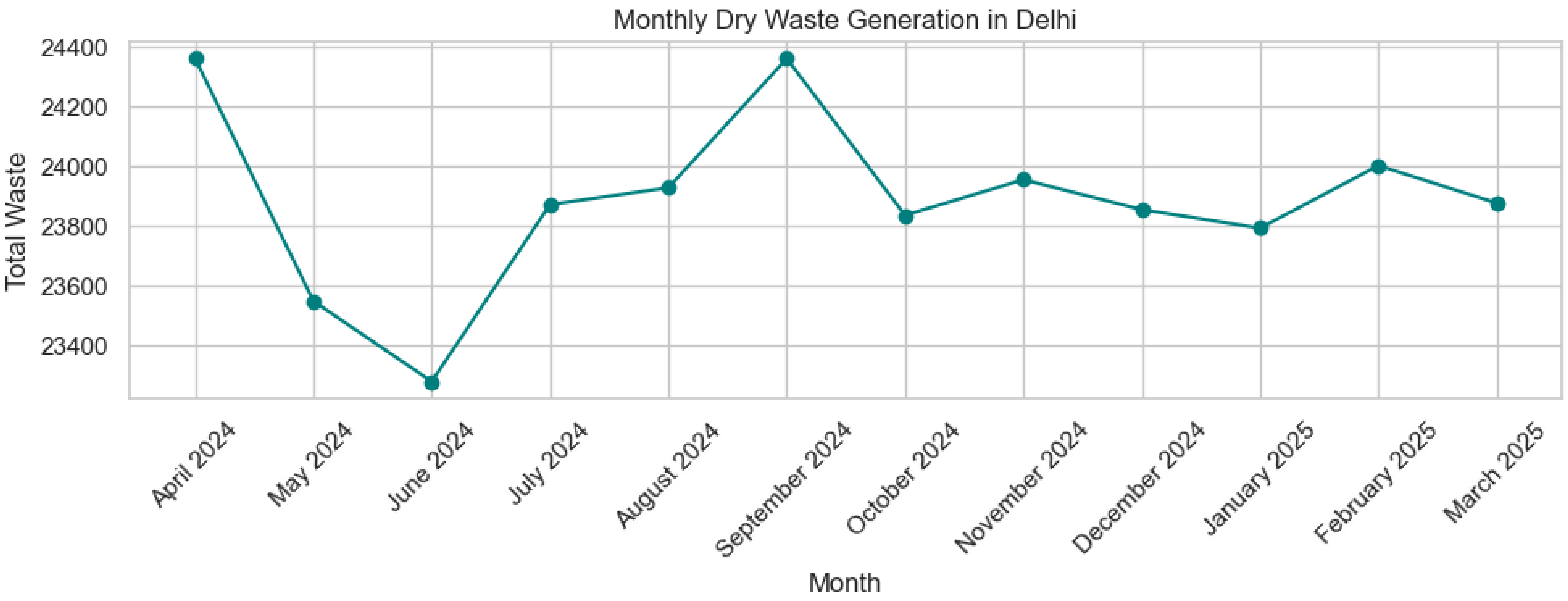


MONTHLY TRENDS IN CITY-WISE WASTE GENERATION CHENNAI

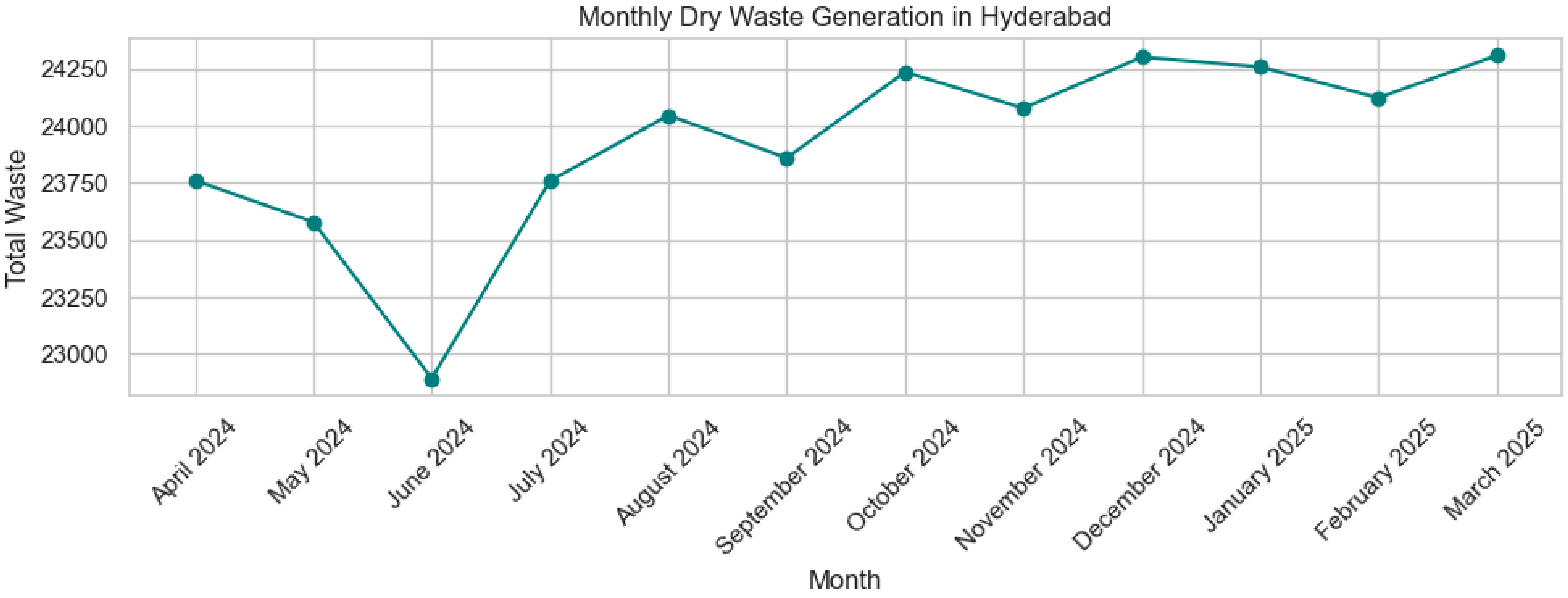


MONTHLY TRENDS IN CITY-WISE WASTE GENERATION

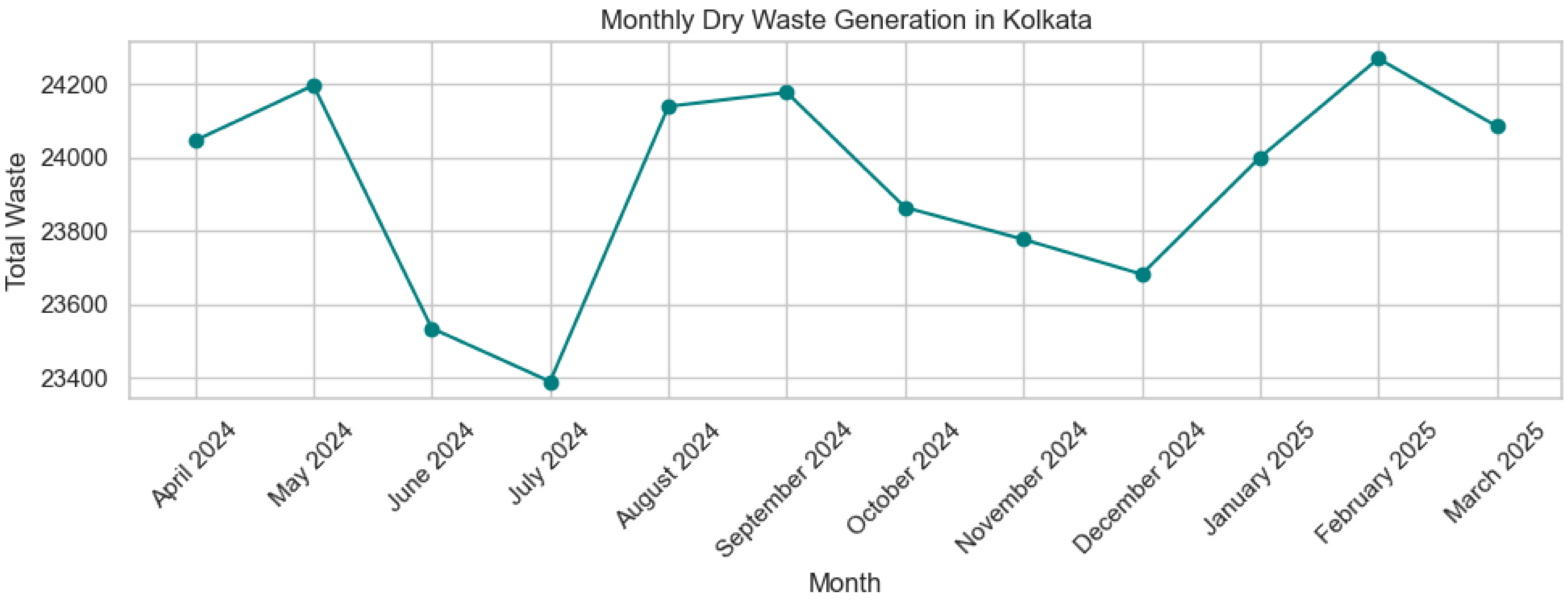
DELHI



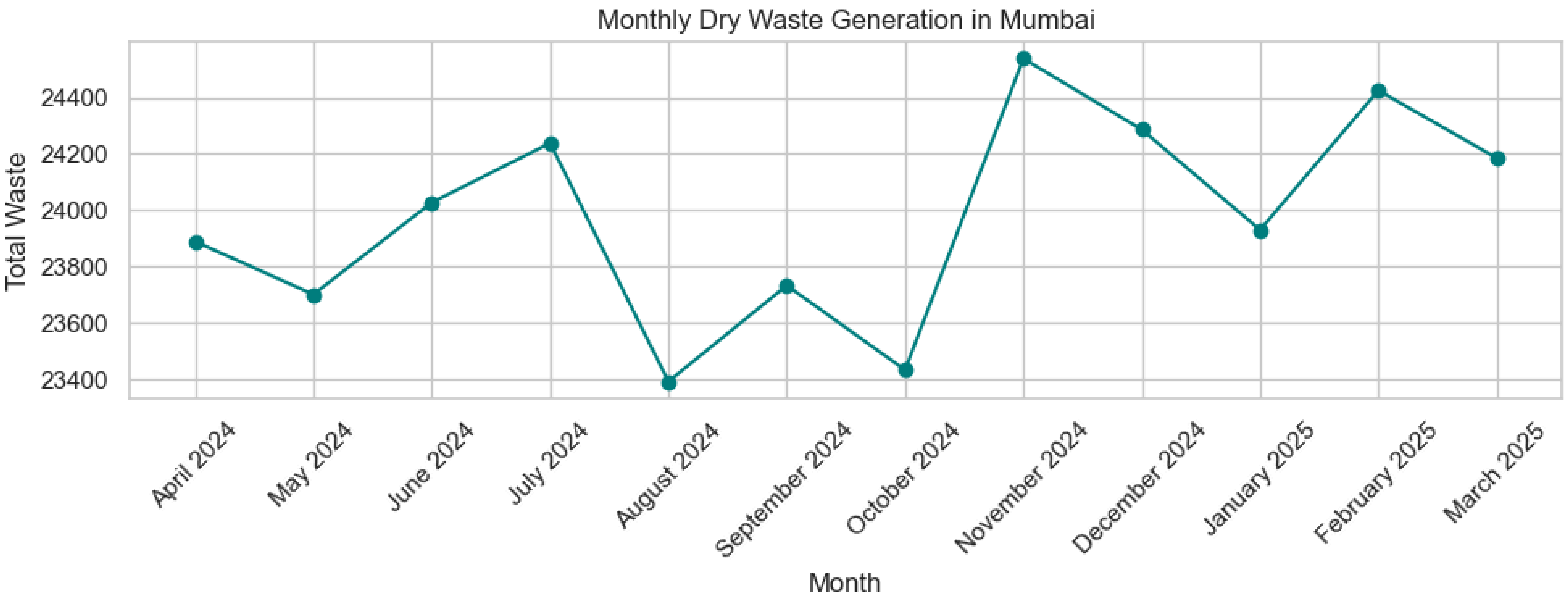
MONTHLY TRENDS IN CITY-WISE WASTE GENERATION HYDERABAD



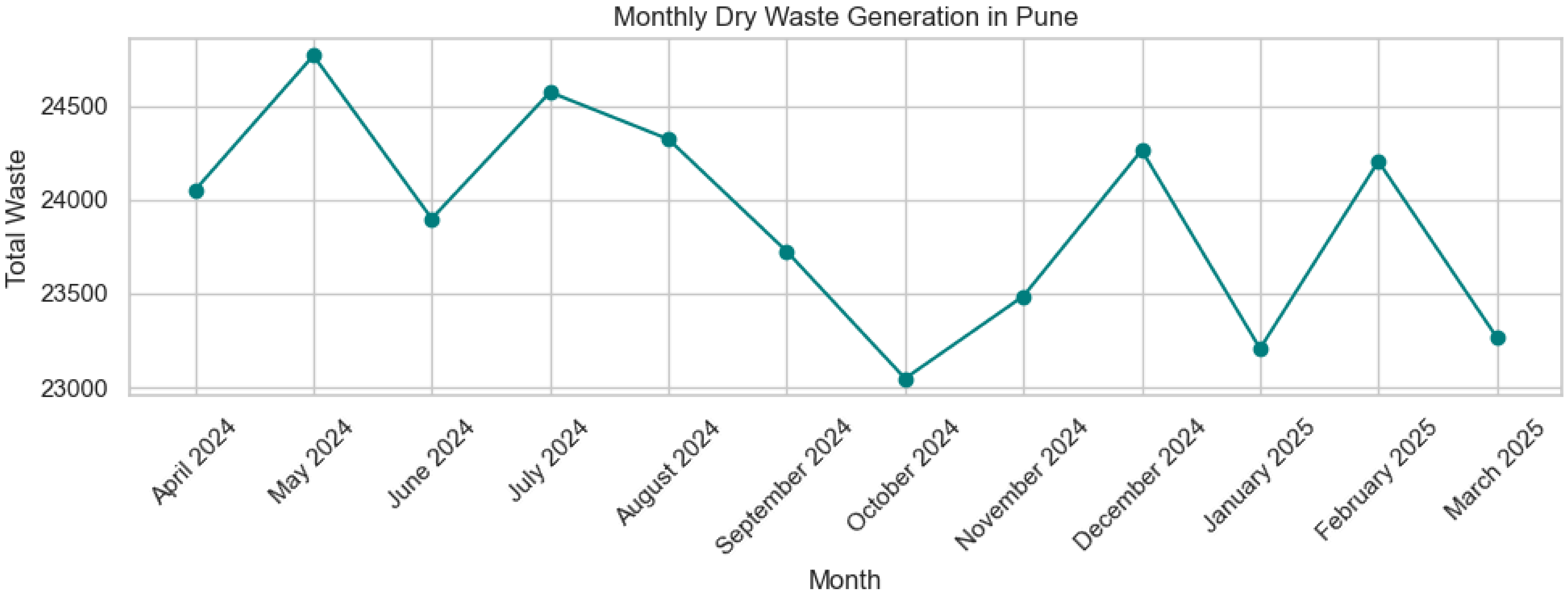
MONTHLY TRENDS IN CITY-WISE WASTE GENERATION KOLKATA



MONTHLY TRENDS IN CITY-WISE WASTE GENERATION MUMBAI



MONTHLY TRENDS IN CITY-WISE WASTE GENERATION PUNE

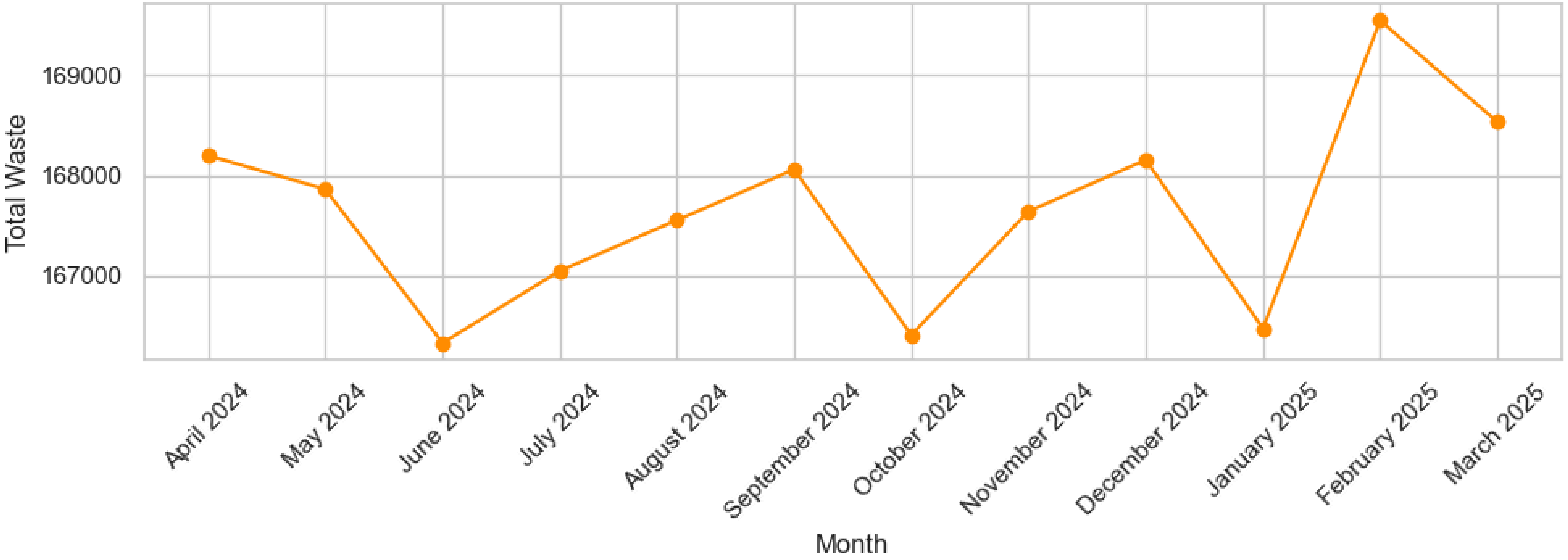


INFERENCE FROM THE PREVIOS GRAPHS

- **Bangalore** has the highest waste generation, but with frequent fluctuations suggesting **seasonal or collection issues.**
- **Chennai and Mumbai** show high volumes, but irregular trends hint at **inconsistent collection or local disruptions.**
- **Pune's** trend is **volatile**, pointing to possible reporting gaps or system inefficiencies.
- **Delhi and Hyderabad** are more stable, which could reflect **efficient systems or a lack of granular data.**
- Waste dips from June to August in most cities, likely due to monsoon-related disruptions.
- **Spikes in February–March** may indicate year-end cleanups or backlog clearances.
- These patterns highlight opportunities for forecasting tools and city-specific planning.

TOTAL MONTHLY DRY WASTE GENERATION ACROSS ALL CITIES

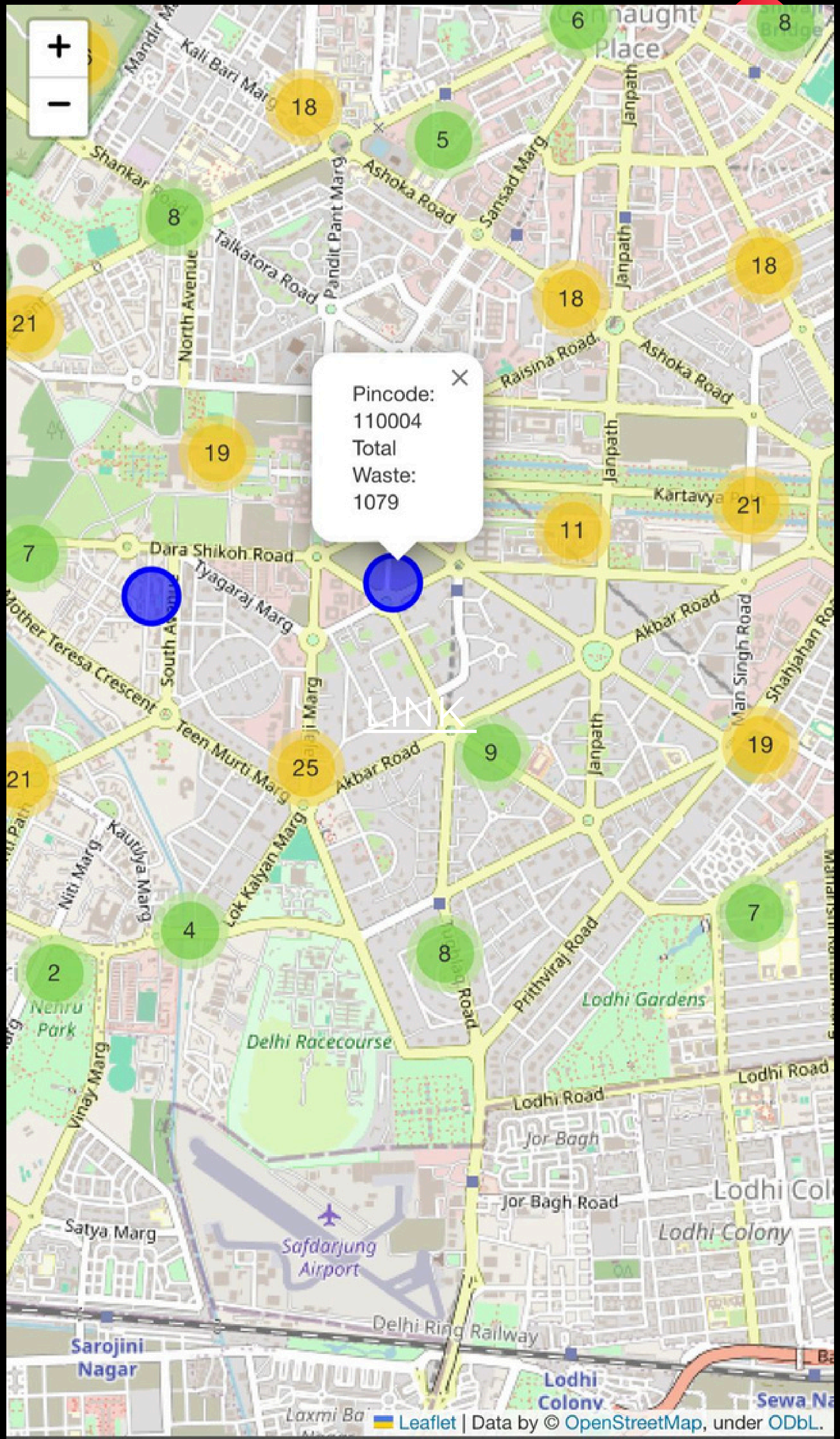
- Waste levels are mostly stable across months.
- Dips in June, October, and January likely due to monsoon or holidays.
- February shows a sharp spike, suggesting year-end cleanups or backlog clearances.
- Indicates predictable seasonal trends Bintix can plan around.



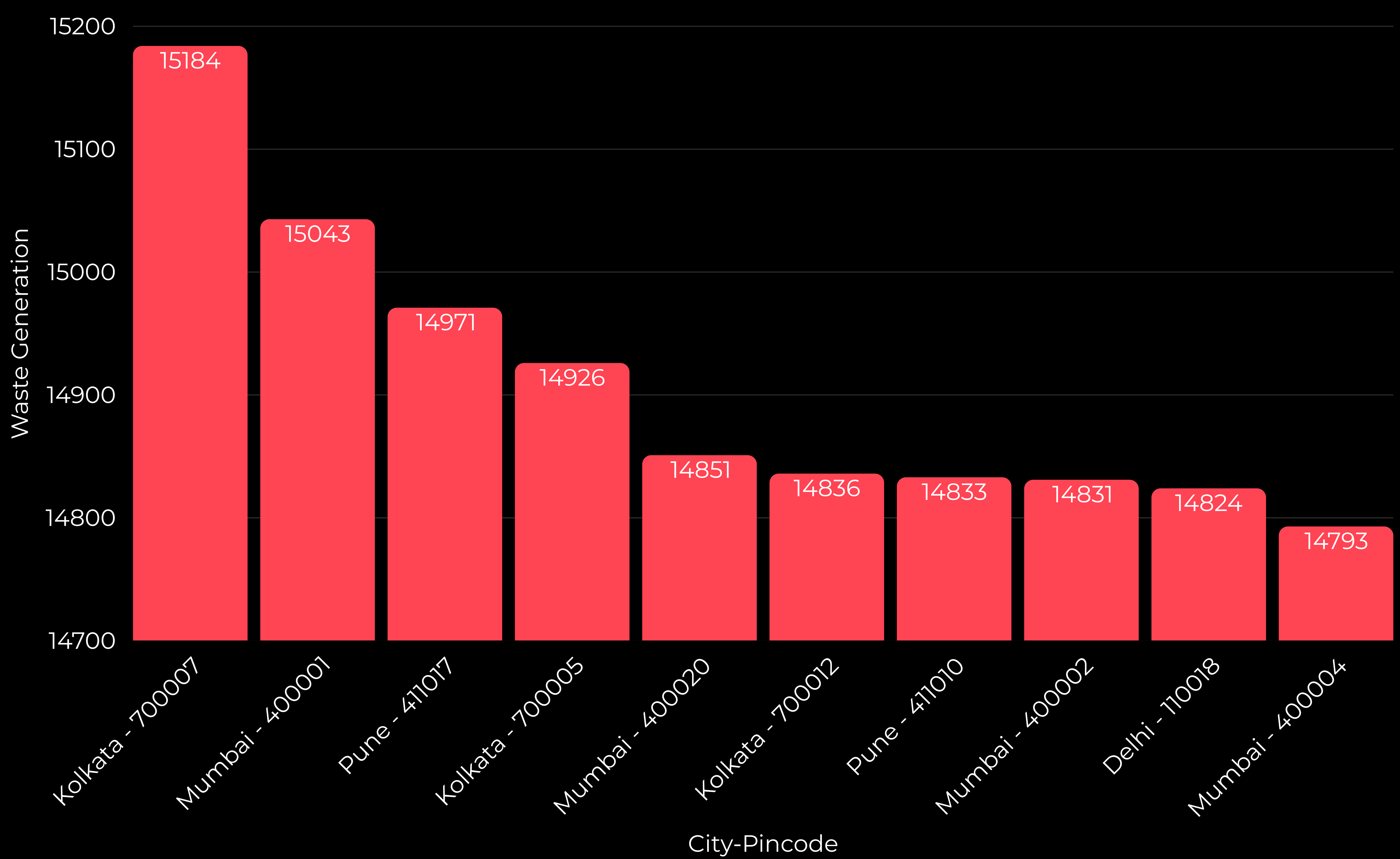
INTERACTIVE HEATMAP - WASTE GENERATION BY PINCODE

To view the interactive map, click here- ***LINK***

- Shows dry waste generation at the pincode level for greater detail within cities.
- Higher waste seen in densely populated and commercial areas.
- Lower waste in peripheral or less developed pincodes.
- Helps identify local waste hotspots within cities.
- Useful for designing targeted and efficient waste management plans.



TOP 10 PINCODES BY WASTE GENERATION



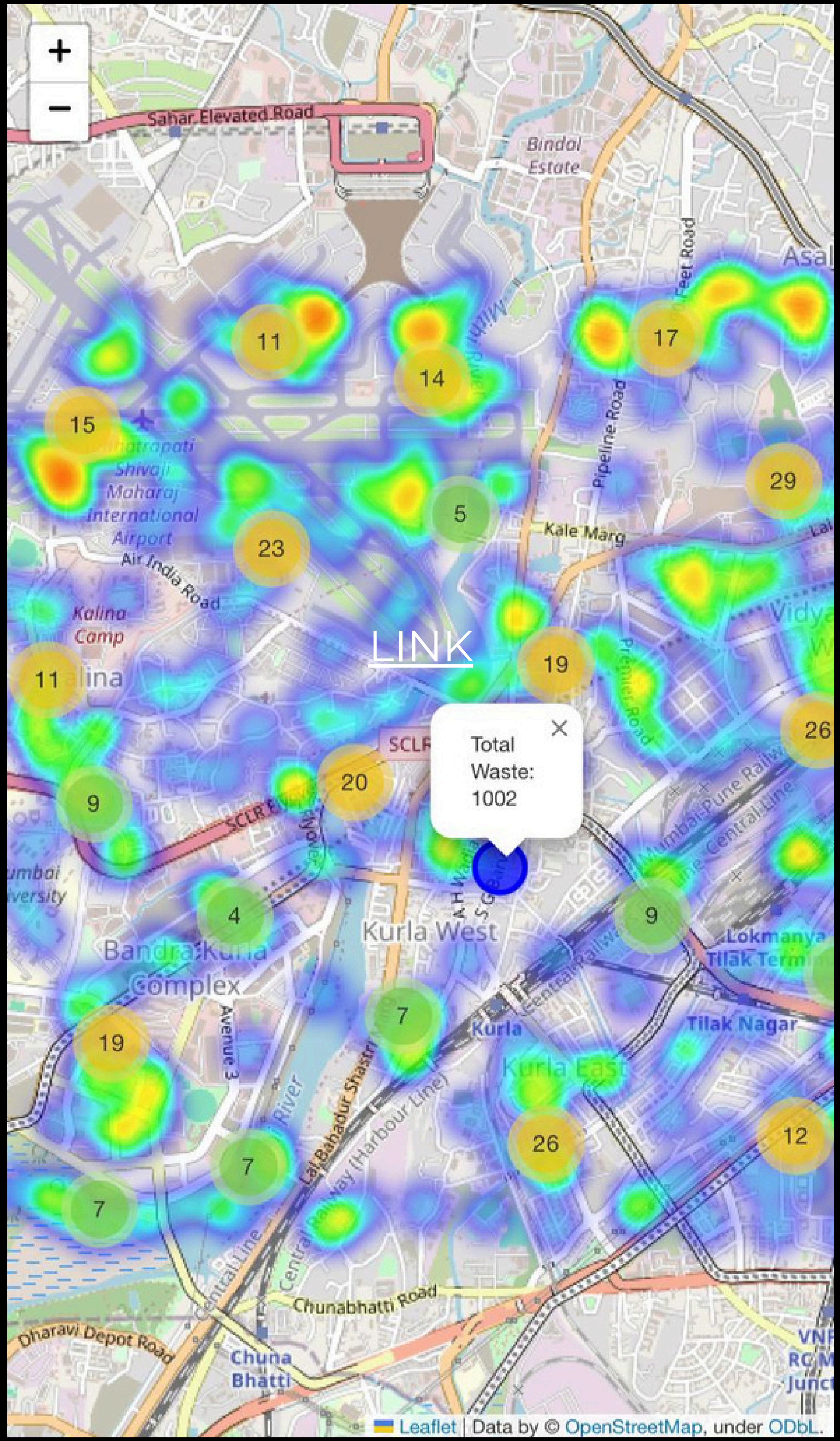
INFERENCE FROM THE PREVIOS GRAPS

- Pincodes from Kolkata, Mumbai, Pune, and Delhi make up the top 10.
- Kolkata’s 700007 leads the list — possibly due to high commercial density.
- Mumbai appears the most, showing how waste is spread across multiple localities.
- No city dominates entirely, which suggests that urban waste hotspots exist across regions.
- These areas can be good starting points for targeted waste management or awareness efforts.

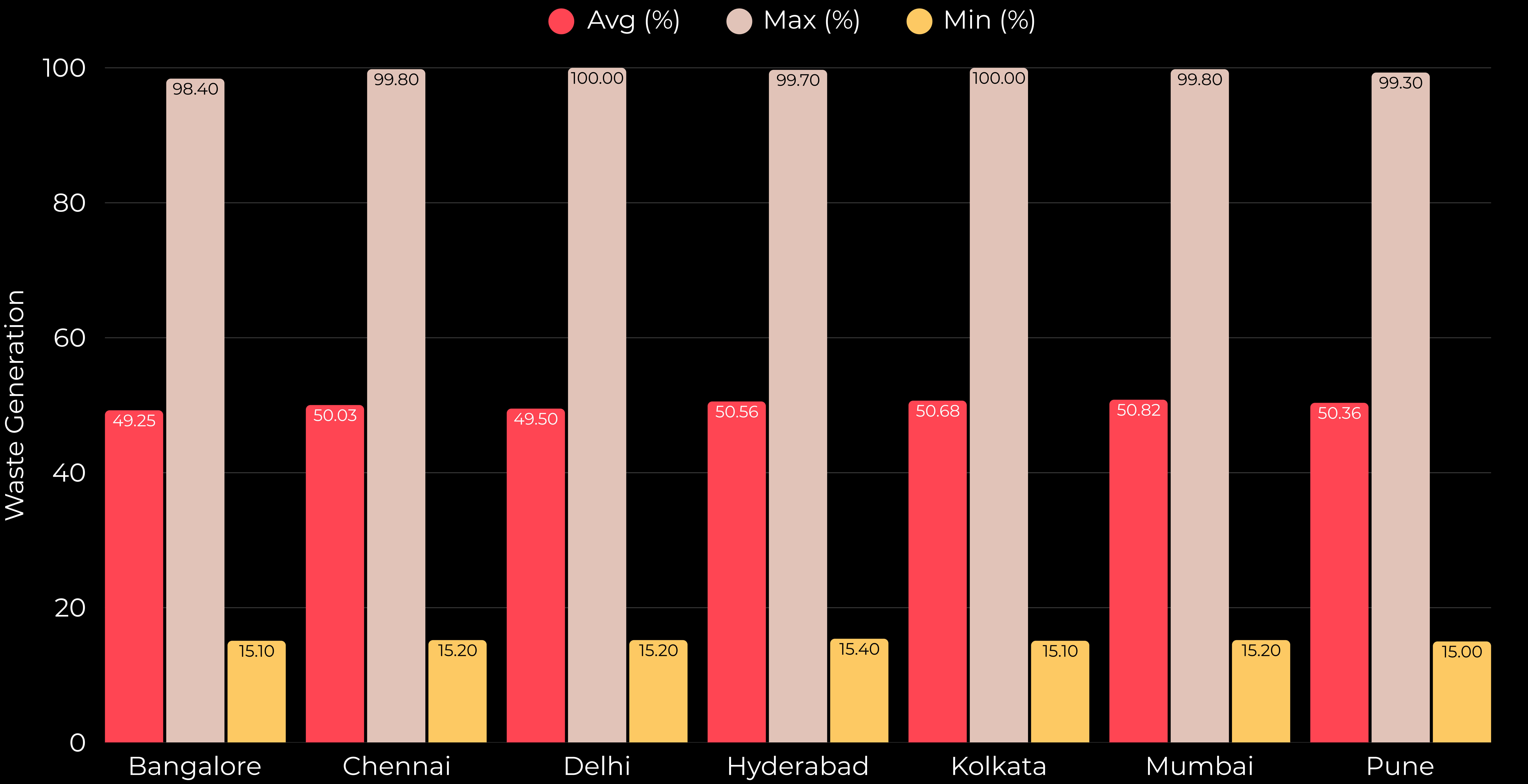
INTERACTIVE HEATMAP - WASTE GENERATION BY COORDINATES

To view the interactive map, click here- ***LINK***

- Displays waste data at the exact latitude and longitude level.
- Reveals fine-grained spatial patterns within each pincode and city.
- Useful for spotting specific buildings, streets, or clusters with high waste.
- Ideal for micro-level planning, such as optimizing bin placement or pickup frequency.
- Offers the most accurate view of spatial waste distribution in the dataset.



PARTICIPATION LEVELS BY CITY: AVERAGE, MINIMUM & MAXIMUM



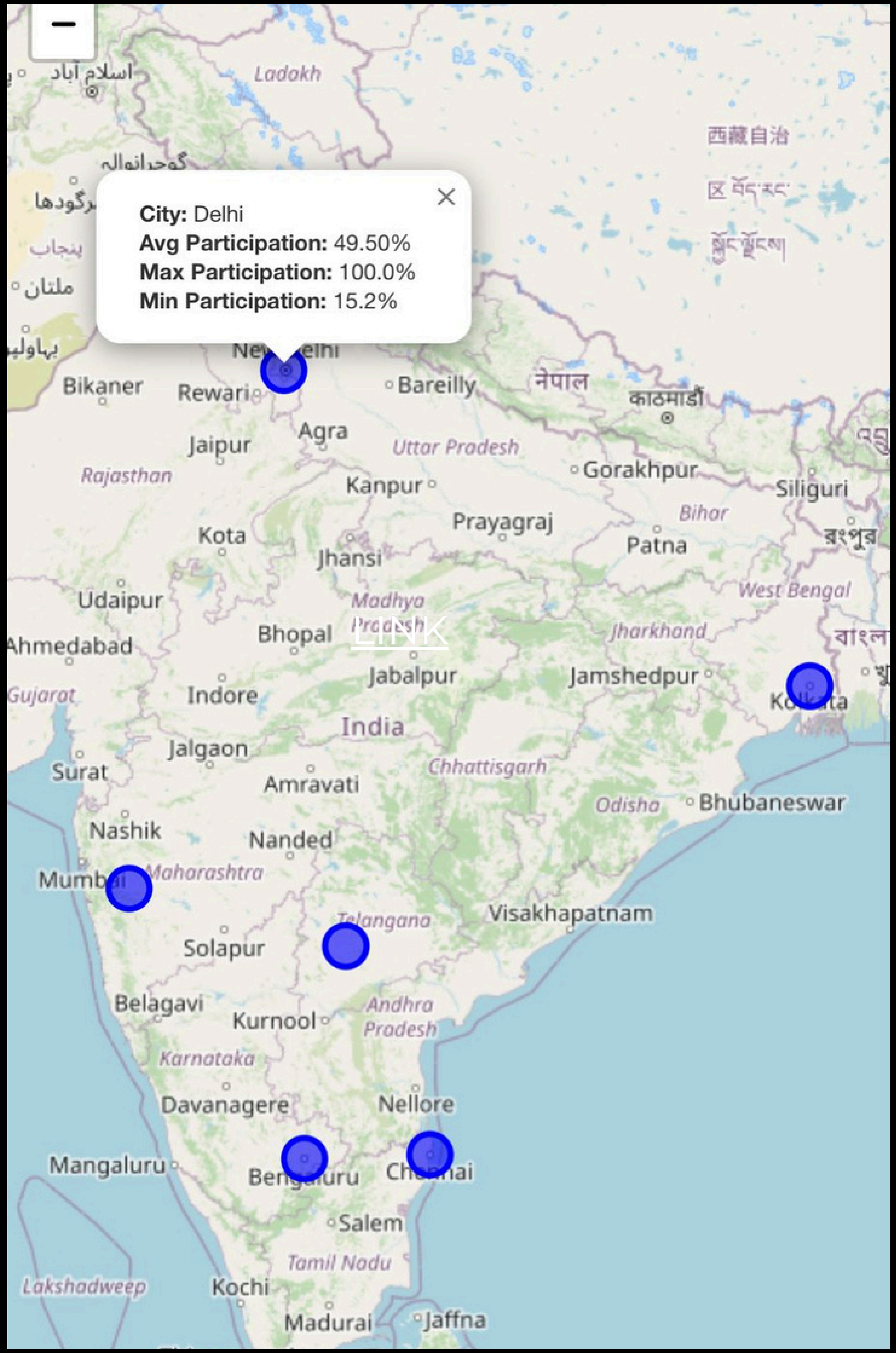
INFERENCE FROM THE PREVIOS GRAPH

- Max participation is nearly 100% in all cities, showing strong potential when systems work well.
- Average participation is around 50%, highlighting a major gap between ideal and actual engagement.
- Minimum participation stays low (~15%), pointing to underperforming zones that need targeted attention.
- Cities like Mumbai, Hyderabad, and Kolkata have slightly better averages, making them good candidates for scaling community initiatives.

PARTICIPATION PERCENTAGE BY CITY

To view the interactive map, click here- ***LINK***

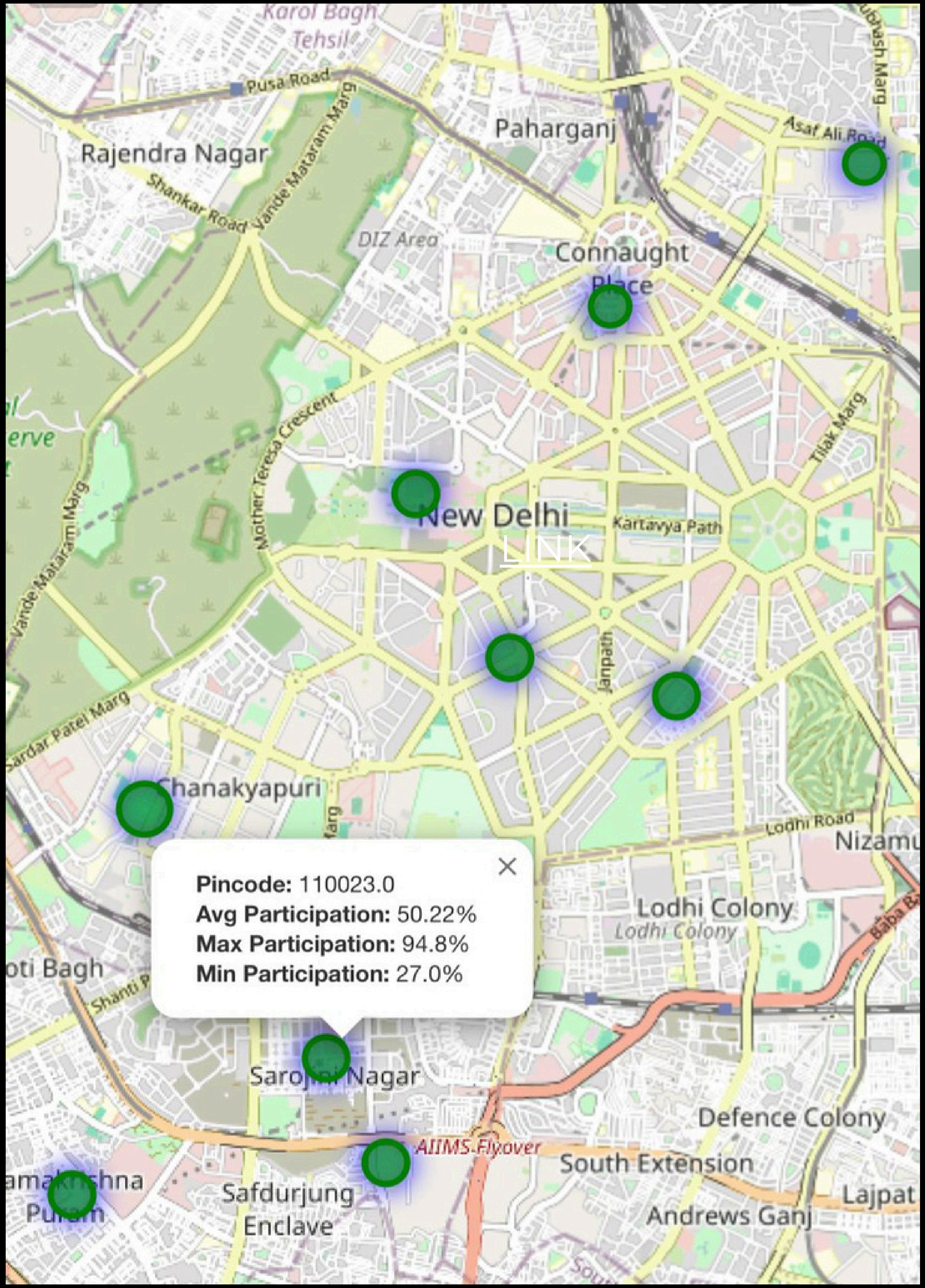
- Shows the average, highest, and lowest participation percentages for each city.
- Helps assess how actively citizens are contributing to waste segregation and disposal efforts.
- Cities with high max but low average may have uneven participation across areas.
- Consistently high averages suggest broader awareness and engagement.



PARTICIPATION PERCENTAGE BY PINCODE

To view the interactive map, click here-
LINK

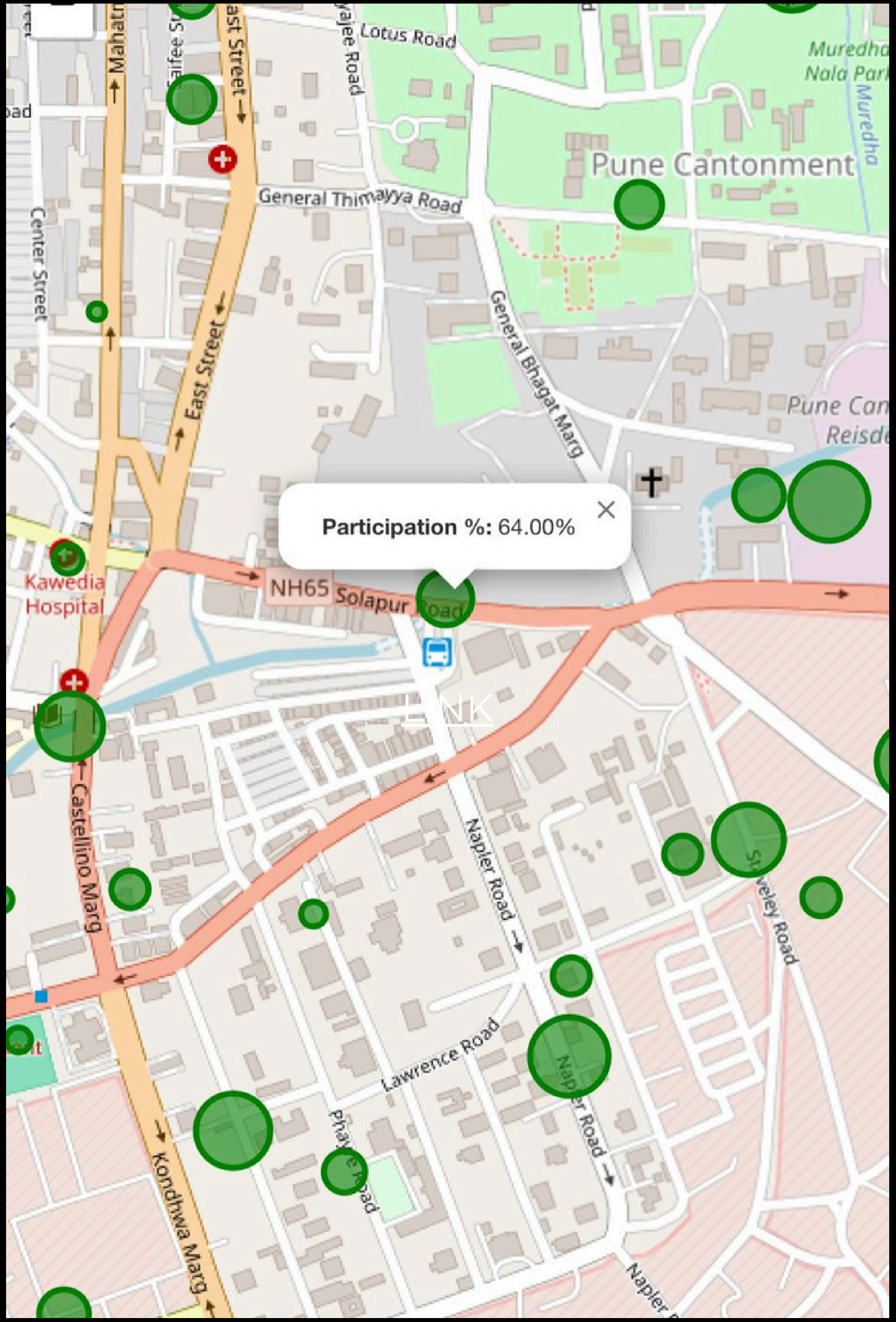
- This map displays the participation percentage across various pincodes.
- Variation in average, maximum, and minimum participation is visible at the local level.
- Highlights how engagement differs within a city.



PARTICIPATION PERCENTAGE BY COORDINATES

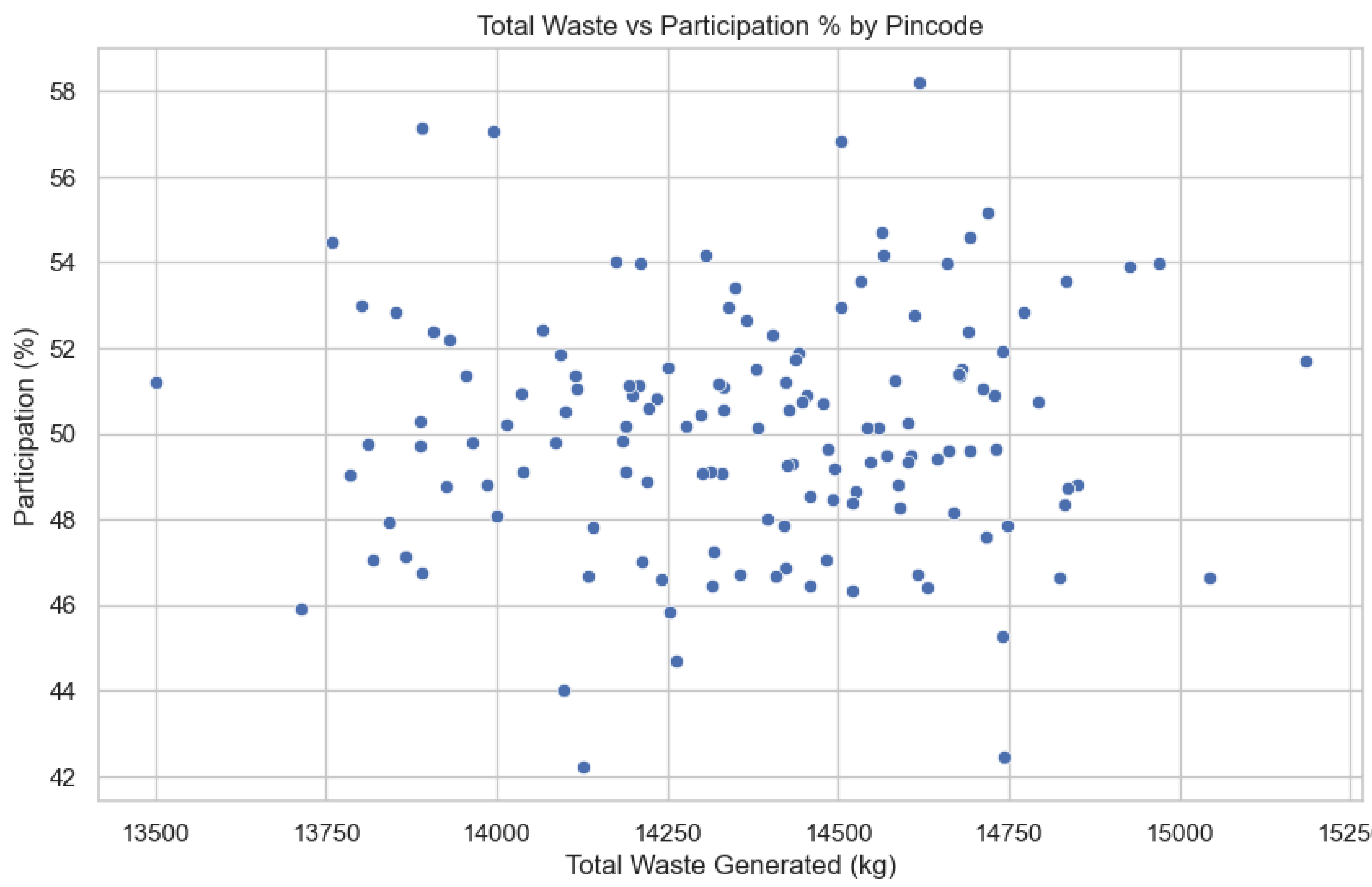
To view the interactive map, click here- ***LINK***

- This map displays participation at the most granular level — based on exact coordinates.
- The size of each circle represents the participation percentage at that location.
- Larger circles indicate higher participation, while smaller ones reflect lower involvement.
- Useful for identifying specific streets or localities with strong or weak community engagement.



SCATTER PLOT OF WASTE GENERATION VS PARTICIPATION %

- This plot uses common pincodes from both datasets.
- No strong correlation between waste generated and participation.
- High waste doesn't always reflect high involvement.
- Most pincodes show moderate values for both, indicating the need to address participation and waste volume separately.

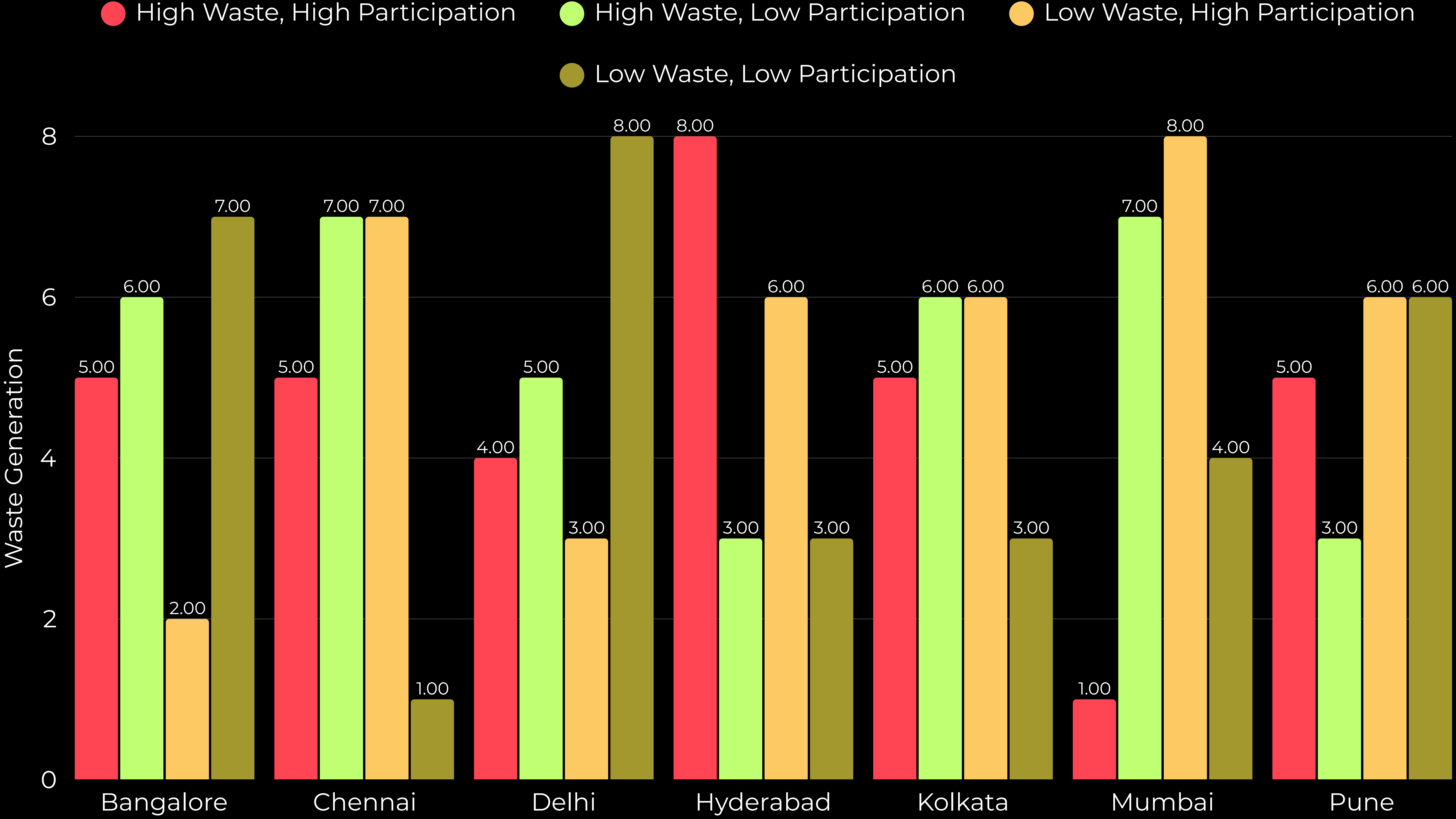


UNDERSTANDING WASTE & PARTICIPATION PATTERNS

To uncover actionable patterns, I classified each pincode into four categories based on whether their total waste and participation % were above or below the median across all pincodes:

1. High Waste, High Participation – Areas generating more waste but also showing strong engagement. Ideal for piloting advanced waste solutions.
2. High Waste, Low Participation – High-impact zones with low involvement. Key targets for awareness and collection drives.
3. Low Waste, High Participation – Smaller waste output but strong participation. Great for identifying best practices.
4. Low Waste, Low Participation – Low output and low involvement. May need educational or access-based strategies.

The next chart shows how many pincodes from each city fall into these categories, helping identify where Bintix could focus different types of interventions.


High Waste, High Participation
High Waste, Low Participation
Low Waste, High Participation
Low Waste, Low Participation
Waste Generation
0
2
4
6
8
Bangalore
Chennai
Delhi
Hyderabad
Kolkata
Mumbai
Pune
5.00
6.00
2.00
7.00
5.00
7.00
7.00
1.00
4.00
5.00
3.00
8.00
8.00
3.00
6.00
3.00
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3.00
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7.00
8.00
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3.00
6.00
6.00

INFERENCE FROM THE PREVIOS GRAPH

- Cities show a mix of categories, reflecting varied waste and participation patterns
- Hyderabad has the most high waste and high participation pincodes, ideal for scaling efforts
- Bangalore and Delhi have several high waste but low participation zones that need focused engagement
- Mumbai and Pune show strong low waste with high participation, useful for identifying best practices
- Low waste and low participation areas, especially in Bangalore and Pune, may need better outreach
- Helps Bintix prioritize actions and replicate effective models


RELATIONSHIP BETWEEN CITY POPULATION AND WASTE GENERATION
Delhi
Bangalore
Mumbai
Kolkata
Pune
Hyderabad
Chennai
288200
288000
287800
287600
287400
287200
287000
286800
286600
0.000
1.000
2.000
3.000
4.000

INFERENCE FROM THE PREVIOS SCATTER PLOT

- The scatter plot shows that cities with larger populations tend to generate more waste, as expected.
- However, the variation is minimal , waste values are closely clustered, even among cities with different population sizes.
- This suggests that waste generation may not scale directly with population alone. Other factors like consumption habits, collection efficiency, and reporting standards may also influence totals.
- Bangalore, despite having a smaller population than Delhi and Mumbai, shows the highest waste generation, indicating possible differences in data coverage or intensity of reporting.

THANK YOU FOR TAKING THE TIME TO REVIEW MY PRESENTATION.
I'M OPEN TO ANY FEEDBACK, SUGGESTIONS, OR CORRECTIONS.
LOOKING FORWARD TO IMPROVING AND LEARNING FURTHER THROUGH YOUR INSIGHTS.

Mail: meenakshipmurali@gmail.com

Contact :9489683941